

Oliver Loo Chong Zheng

Security engineer and product builder. I break things, I build things, and I care how they feel.

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EDUCATION

- National University of Singapore — Bachelor of Computing (Honours), Computer Science**2021 – 2026
- Second Class Upper Honours (Distinction), CAP 4.02 / 5.00. Security specialisation. Degree conferred 31 August 2026.
- Stanford University — Undergraduate Summer Visitor, 8 credits transferred**Summer 2025
- GPA 4.00 / 4.00. Computer Graphics and Imaging (A+), Technology Entrepreneurship (A-).

EXPERIENCE

- DART Consulting and Training — Technology Attachment (NUS ATAP)**2026
- Singapore
- Worked inside a cybersecurity training and consulting environment, deepening practical security knowledge beyond coursework into how security is operated and adopted across an organisation.
 - Developed and delivered technical material and supported training communications, translating security concepts into content non-specialists could learn from and act on.
- HTX, Home Team Science and Technology Agency — AI and Cybersecurity Intern**Jan – Jun 2025
- Singapore
- Built automated data collection and workflow orchestration systems for social media platforms.
 - Deployed and managed lightweight LLMs in virtualised Linux environments, integrating them into data collection and online interaction pipelines.
 - Applied advanced AI techniques including vision-language models, captcha solvers and decision-making algorithms.
- Gem Comm — AI and Digital Marketing Intern**Mar – Jun 2024
- Singapore
- Designed automated workflows on the OpenAI API to streamline social media content generation.
 - Researched and proposed AI solutions for finance, including sentiment analysis of financial news and predictive models for stock movement.
- Singapore Armed Forces, 10C4I — Platoon Sergeant (Second Sergeant)**Aug 2019 – Jul 2020
- Led a team maintaining and troubleshooting a critical army network, configuring and managing network devices, and delivering the training programme that integrated new operators.

PROJECTS

- Levanta — AI personal training app, live on the App Store**2025 – 2026
- Solo-designed and shipped an iOS app in React Native and Expo with local-first storage in WatermelonDB and a Google Gemini coaching layer.
 - Natural-language workout logging parses free text into structured sets; every AI-proposed action renders as a confirmable widget so the model never mutates user data silently.
- funhop.app — Singapore night-planner, live**2026
- Built the full stack in Next.js and TypeScript on Supabase, plus a seven-stage Python ingestion pipeline that scrapes Telegram and publisher sites and uses an LLM to extract structured events.
 - Diagnosed and fixed silent data corruption affecting roughly 85% of the events table, traced to stripped query strings on signed image URLs and synthesised detail URLs returning 404.
- Langfun — Oui Chef! and Sushi Senpai — language-learning arcade games**2026
- Built one reusable game engine in Expo, Skia and Reanimated serving two titles, with all language content isolated to JSON and a theme file so a new title needs no engine changes.
 - Implemented a Leitner spaced-repetition scheduler and plausibility-constrained distractor selection so players learn the phrases rather than pattern-match the options.
- Bao Bae — retrieval-augmented couples therapy platform**2025
- Built a RAG system using FAISS vector search over a therapeutic knowledge base, combining clinical material with real-time conversation analysis to deliver grounded Emotionally Focused Therapy sessions.

SECURITY PRACTICE & CERTIFICATIONS

- Hands-on labs:** Offensive Security Proving Grounds, HackSmarter and Hack The Box. Enumeration, exploitation and privilege escalation across Linux and Windows targets.
- Malware analysis and detection:** completed the *Practical Malware Analysis* lab set (static triage, dynamic analysis in isolated VMs, disassembly), and built a PCAP analysis agent with an MCP server that surfaces suspicious flows into an LLM analyst loop.
- Certifications:** Google Cybersecurity Professional Certificate, AWS Cloud Essentials.

SKILLS

Languages	Python, TypeScript, JavaScript, Java, C, Rust, SQL, Bash
Building	React, React Native, Expo, Next.js, Node.js, Tailwind, Skia, Git, Unix, Docker
Data & AI	Supabase, PostgreSQL, MongoDB, WatermelonDB, FAISS, RAG, LLM integration, Gemini and OpenAI APIs
Security	Penetration testing, malware analysis, network security, vulnerability assessment, Linux hardening